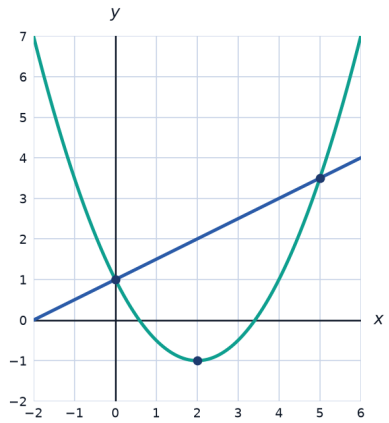
The discriminant and the quadratic formula

Core Concepts: The discriminant and the quadratic formula

1 For the model $f(x) = 3x + 5$ in the discriminant and the quadratic formula, evaluate $f(2)$.



a. 11

b. 14

c. 6

d. 15

2 Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

3 A data set recorded for the discriminant and the quadratic formula consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 3x + 5$ in the discriminant and the quadratic formula, evaluate $f(3)$.

a. 14

b. 17

c. 9

d. 18

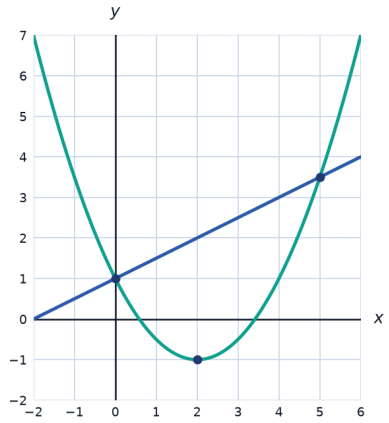
5 Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$



6 A data set recorded for the discriminant and the quadratic formula consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

Graphical & Geometric Interpretation

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7 For the model $f(x) = 3x + 5$ in the discriminant and the quadratic formula, evaluate $f(4)$.

a. 17

b. 20

c. 12

d. 21

8 Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

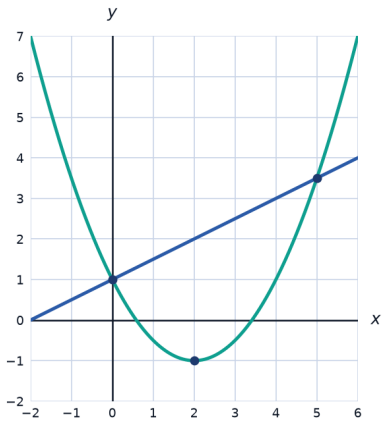
9 A data set recorded for the discriminant and the quadratic formula consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

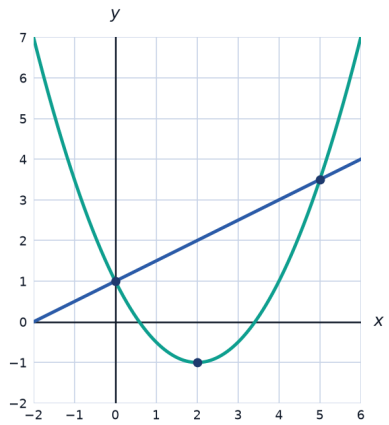
d. 25.5



Exam-Style Problem Solving & Modeling

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10 For the model $f(x) = 3x + 5$ in the discriminant and the quadratic formula, evaluate $f(5)$.



a. 20

b. 23

c. 15

d. 24

11 Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

12 A data set recorded for the discriminant and the quadratic formula consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

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